IDC 4140 – 0001

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Project 3 - Reflection

1. Introduction

The objective of this project is to develop and evaluate machine learning models capable of classifying complex medical images. Specifically, this project utilizes the DermaMNIST dataset to categorize images of skin lesions into one of 7 distinct diagnostic classes. Because medical imagery is highly complex and non-linear, this report compares the performance of a baseline linear model (Logistic Regression) against a non-linear Multi-Layer Perceptron (Neural Network) to determine the best approach for this classification task.

2. Dataset Description & Exploratory Data Analysis

The DermaMNIST dataset consists of 28×28 pixel RGB images. The data is pre-split into three subsets: 7,007 training images, 1,003 validation images, and 2,005 test images.

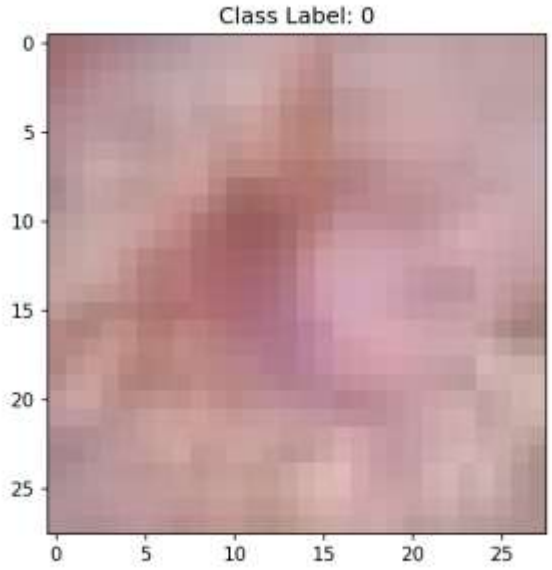

Figure 1: Sample 28 x 28 skin lesion image from the training set.

An analysis of the training and validation sets revealed a severe class imbalance. As shown in Figure 2, Class 5 dominates the dataset. For context, in the validation set of 1,003 images, 671 belong to Class 5, while minority classes (such as Class 3) contain as few as 12 images.

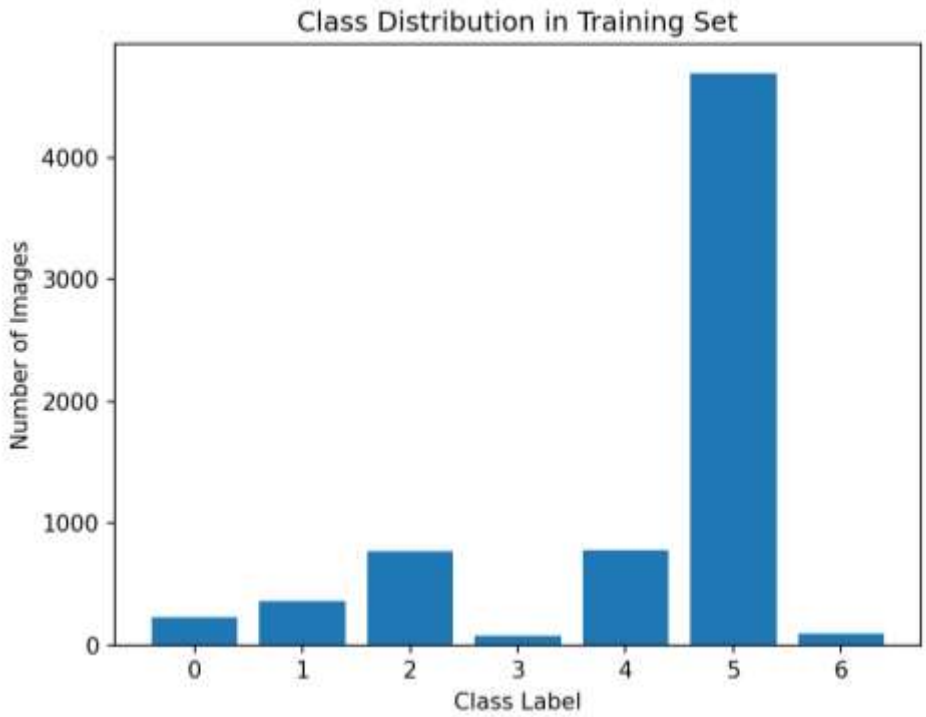

Figure 2: Distribution of classes within the training dataset showing severe imbalance toward Class 5.

3. Preprocessing and Methodology

To prepare the image data for the models, two critical preprocessing steps were applied:

1. Normalization: The pixel values, which originally range from 0 to 255, were divided by 255.0 to scale them between 0.0 and 1.0. This ensures faster mathematical convergence during training.
2. Flattening: Because Scikit-Learn models require 2D arrays (samples $\times$ features), the 3D image arrays ($28 \times 28 \times 3$) were reshaped into a 1D vector of 2,352 features per image.

4. Baseline Model: Logistic Regression

To establish a performance baseline, a Logistic Regression model was initialized. Logistic Regression is fundamentally a linear classifier; it works by attempting to draw a straight, multidimensional mathematical plane (a hyperplane) through the dataset to separate the different

categories. Because the original 28×28 RGB images were flattened, the model was forced to draw this linear boundary across a highly complex, 2,352-dimensional feature space.

Results and Behavior:

The model was trained with max_iter=3000 to allow the optimization algorithm ample time to converge. It achieved a Validation Accuracy of 68.99% and a Test Accuracy of 69.63%.

While this accuracy initially appears adequate, evaluating the model's behavior through the classification report and confusion matrix reveals a severe bias known as the "Accuracy Paradox." Because medical images of skin lesions are organic, complex, and highly non-linear, a straight mathematical line is entirely insufficient to separate them. Consequently, the model failed to find distinct boundaries for the rare diseases and instead optimized its accuracy by overwhelmingly defaulting to the majority class.

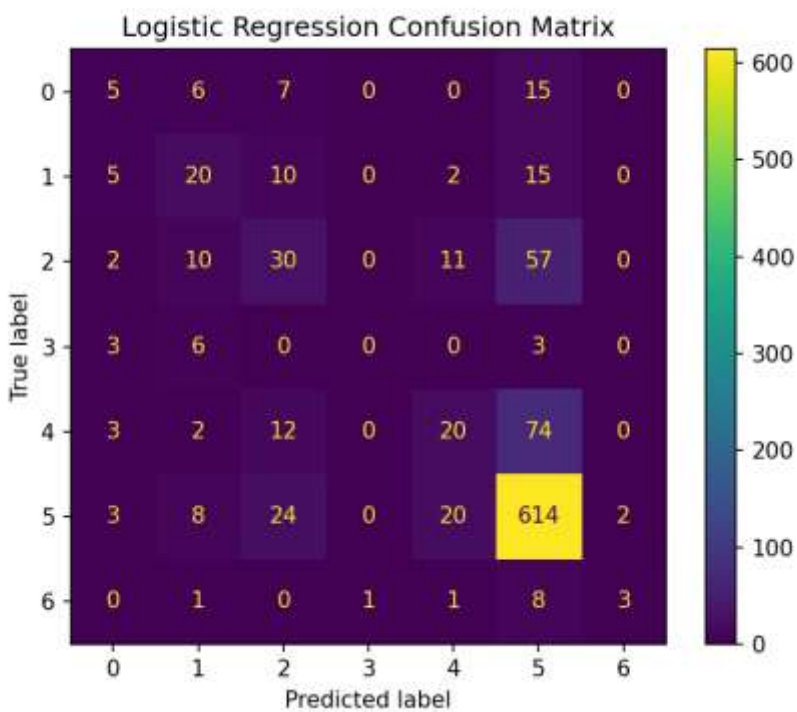

Figure 3: Confusion matrix for the Logistic Regression model. Note the heavy concentration of predictions strictly in the Class 5 column.

This behavior is quantitatively proven by the F1-scores. The model achieved a relatively strong F1-score of 0.84 on Class 5. However, it failed entirely on minority classes, achieving an F1-score of exactly 0.00 on Class 3 (0 Precision, 0 Recall). The confusion matrix visually confirms

this, showing that when the true label was anything other than 5, the model frequently misclassified it as Class 5 simply because it was the most statistically safe "guess."

5. Neural Network Model (MLPClassifier)

Because the linear baseline failed to capture the complexity of the medical images, a Multi-Layer Perceptron (Neural Network) was implemented. Unlike Logistic Regression, a neural network is capable of learning complex, non-linear relationships. It achieves this by passing the input data through multiple layers of artificial neurons and using backpropagation to continuously adjust internal weights based on its errors.

For this dataset, the network architecture was designed with two hidden layers containing 128 and 64 neurons, respectively. Crucially, it utilized the Rectified Linear Unit ('relu') activation function. The ReLU function introduces non-linearity into the network, allowing the mathematical boundaries to bend and warp around the complex pixel patterns (such as lesion borders or color gradients) that the straight lines of Logistic Regression missed.

This architectural shift yielded immediate improvements. The Neural Network increased the overall Validation Accuracy to 71%. More importantly, the hidden layers successfully extracted deeper graphical patterns for some of the minority classes. For example, the F1-score for Class 1 improved from 0.38 in the linear baseline to 0.48 in the neural network, proving that the non-linear approach is much better suited for image classification.

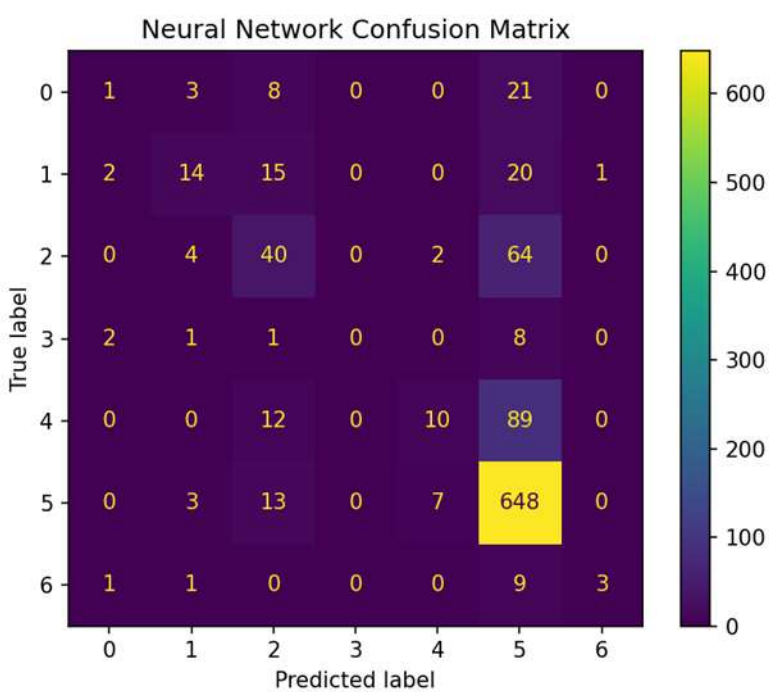

Figure 4: Confusion matrix for the Neural Network model. The diagonal shows slight improvements in correct predictions for minority classes compared to the linear model.

Despite these improvements, the neural network still struggled with the dataset's extreme imbalance. During evaluation, Scikit-Learn triggered an UndefinedMetricWarning for Precision. Precision is mathematically defined as True (True Positives/Positives+False Positives). Because classes like Class 3 contained only 12 images, the network did not have enough data to adjust its weights for that specific disease. As a result, it predicted Class 3 exactly zero times, resulting a divide-by-zero error in the precision formula. This behavior demonstrates a critical machine learning concept: while a neural network's architecture is vastly superior to a linear model for image data, changing the model cannot magically resolve a fundamental lack of training data.6. Discussion and Conclusion

Comparing the two models reveals a clear trade-off. The Neural Network ultimately outperformed the Logistic Regression model in both overall accuracy (71% vs 68.99%) and in identifying specific minority classes (Class 1). However, the Neural Network required significantly more computational power and training time due to the complex backpropagation through its hidden layers.

To resolve the model's inability to predict minority classes (which caused undefined precision metrics), future iterations of this project should implement techniques to balance the dataset. This could include Data Augmentation (artificially creating more images of the rare classes by rotating or flipping existing ones) or using algorithms like SMOTE (Synthetic Minority Over-sampling Technique) to synthetically generate data for the underrepresented skin lesions.

Furthermore, this project highlights the severe limitations of training machine learning models on imbalanced medical data. While transitioning from a linear model to a non-linear Neural Network improved overall performance, changing the architecture alone did not magically solve the underlying class imbalance. Future improvements to this pipeline should include data augmentation techniques (like rotating or flipping the rare images) to artificially balance the dataset before training.

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During the preparation of this project, the author used Google's Gemini to assist with troubleshooting Scikit-Learn code, structuring the report, and refining explanations of evaluation metrics. After using this tool, the author reviewed, edited, and validated all generated code and text, and takes full responsibility for the final content of this report.